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Impact of Collection Method and Storage Time on LDH Reliability

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Introduction

As a near-universal cytosolic enzyme underpinning cellular energy metabolism, lactate dehydrogenase (LDH) is also exceptionally susceptible to pre-analytical artifact. Its measured levels shaped as much by sample handling and storage as by underlying physiology.

Objective

The aim of this study was to evaluate the influence of blood collection tube type on LDH measurement and to assess the effect of sample storage on this variation.

Materials and methods

This was a prospective, analytical, comparative study conducted at the biochemistry laboratory of the Pierre and Marie Curie Center in Algiers. One hundred paired blood samples, collected simultaneously into plain tubes without a clot activator and into lithium-heparin tubes, were analyzed using the cobas c 501 automated analyzer. Measurements were performed immediately after centrifugation and repeated after sample storage at +4 °C for 24 hours.

Results and discussion

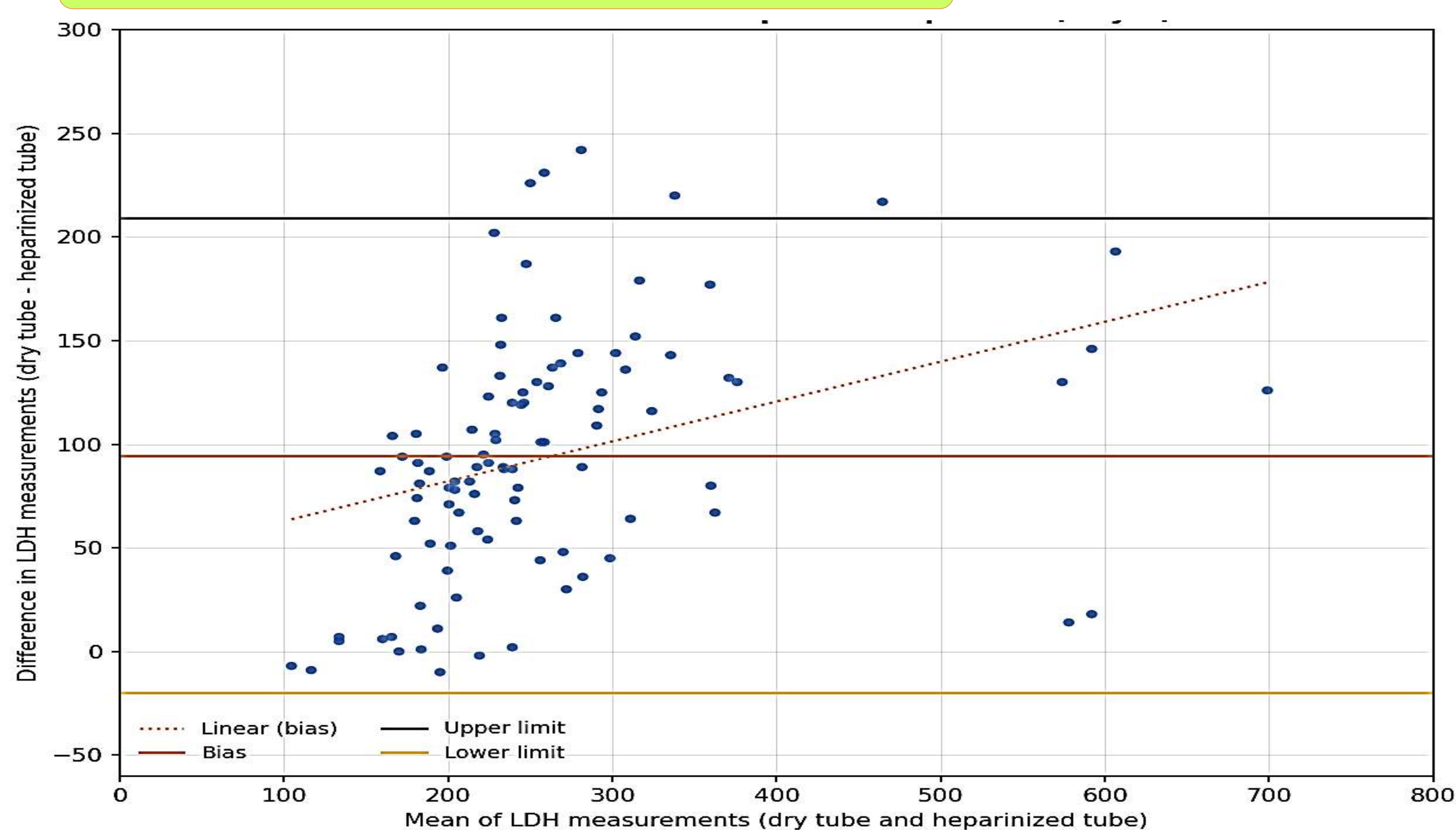


Figure 1:BLAND ALTMAN COMPARING LDH VALUES BETWEEN SERUM AND HEPARINIZED PLASMA

LDH values were significantly higher in serum than in heparinized plasma, with a mean increase of approximately 29% and a mean bias of +95 IU/L ($p < 0.001$).

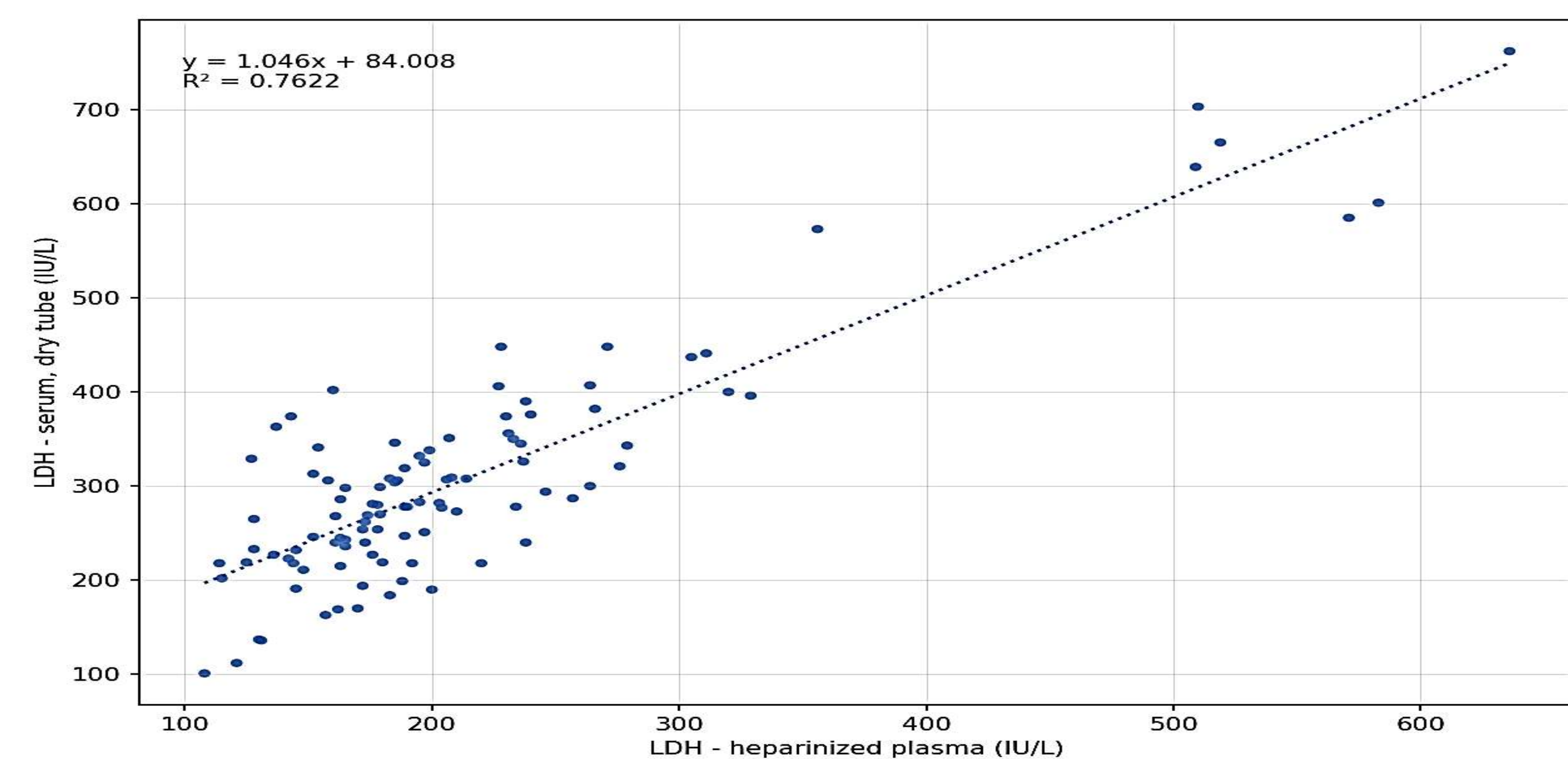


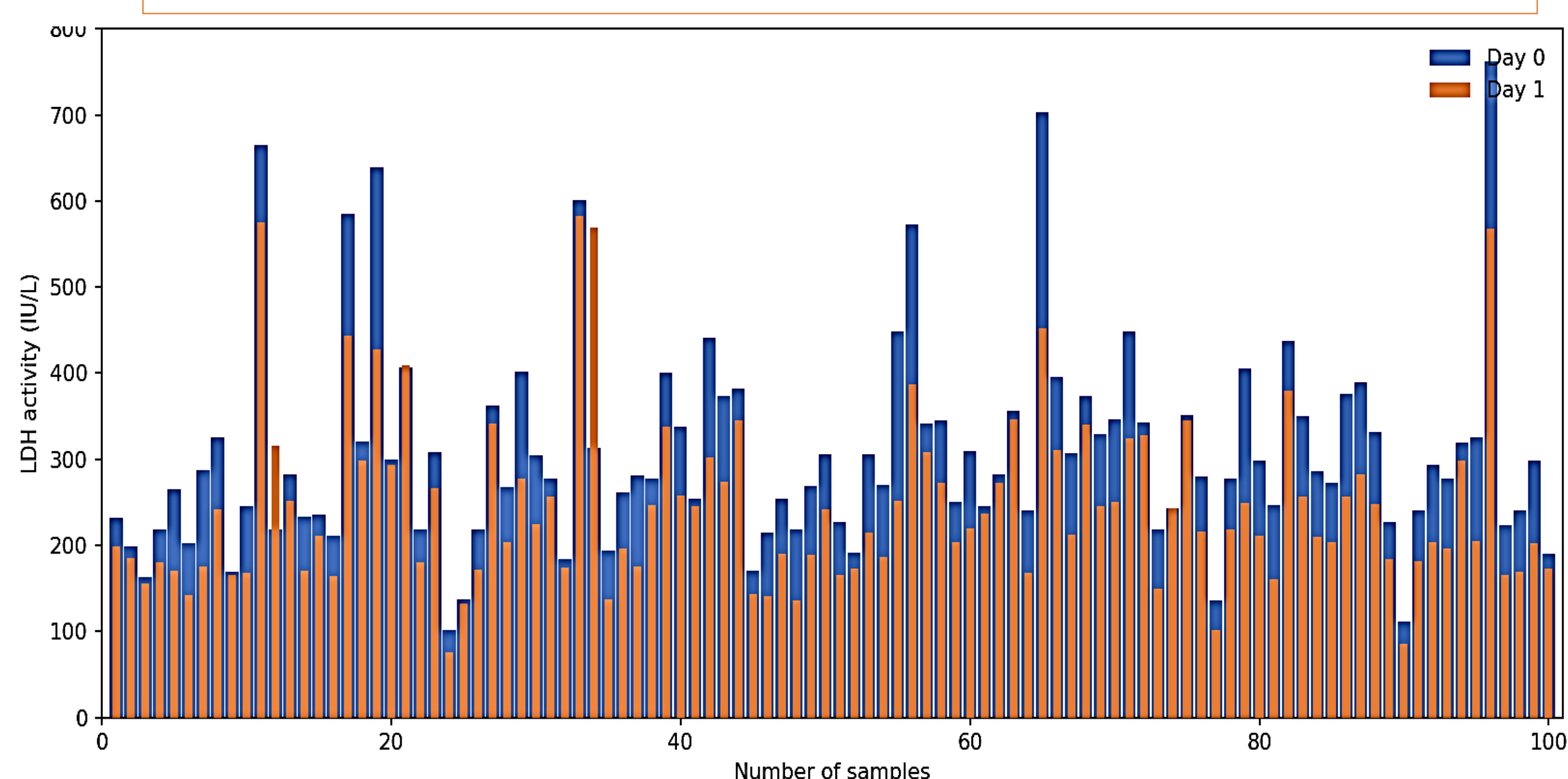
Figure 2:LINEAR REGRESSION BETWEEN LDH VALUES IN SERUM AND HEPARINIZED PLASMA

A very good correlation was observed between the two matrices ($r = 0.873$); however, agreement analyses revealed a systematic bias limiting their interchangeable use.

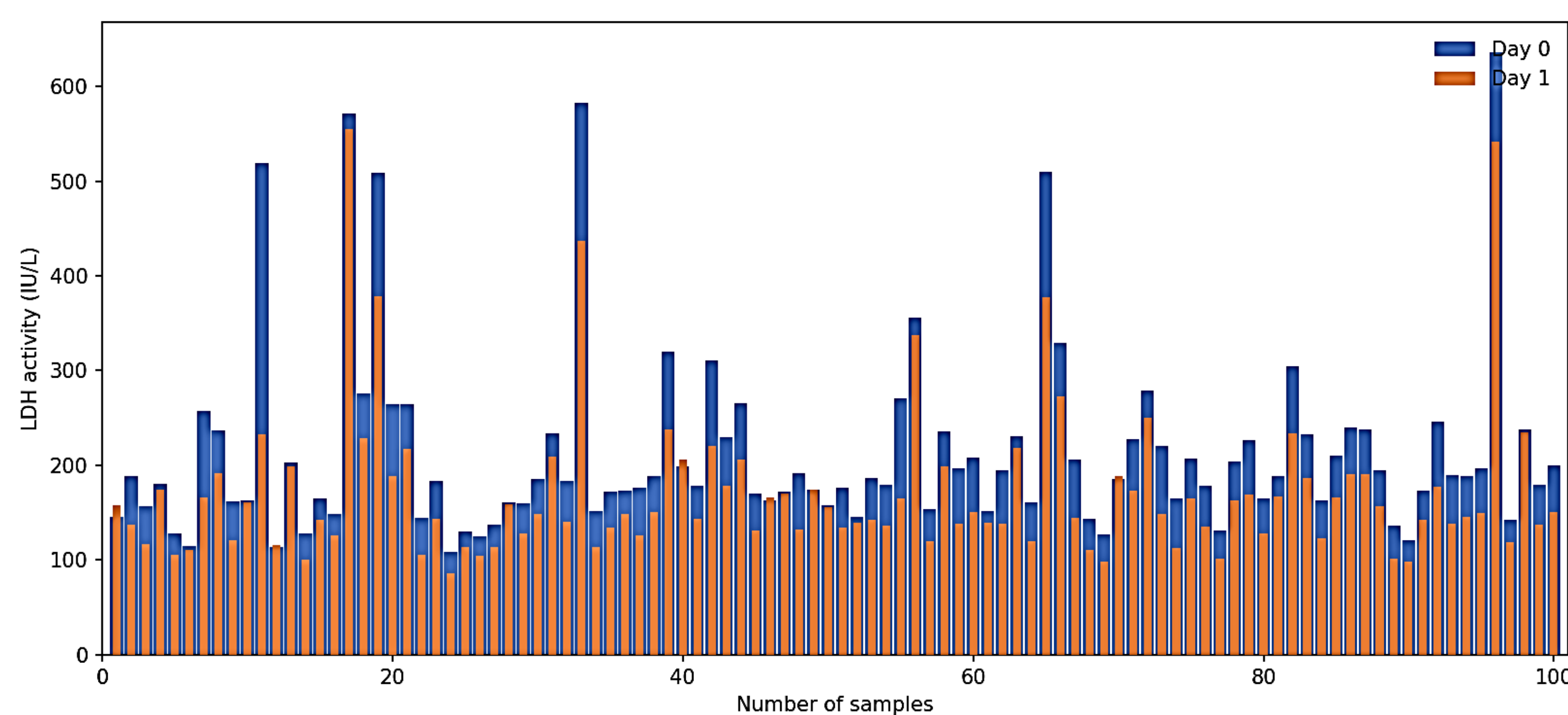
Bibliographic reference

- 1) Hedayati, M., Razavi, S. A., Boroomand, S., & Kheradmand Kia, S. (2020). The impact of pre-analytical variations on biochemical analytes stability : A systematic review. Journal of Clinical Laboratory Analysis, 34(12), e23551. <https://doi.org/10.1002/jcla.23551>
- 2) Rasmussen, M. S., & Pedersen, L. (2024). A new assay falsely increases lactate dehydrogenase in plasma but not in serum. Clinical Biochemistry, 131-132, 110804. <https://doi.org/10.1016/j.clinbiochem.2024.110804>

Figure 3: DISTRIBUTION OF LDH VALUES BEFORE AND AFTER SOTRAGE



3A: DISTRIBUTION OF LDH VALUES BEFORE AND AFTER SOTRAGE IN SERUM



3B: DISTRIBUTION OF LDH VALUES BEFORE AND AFTER SOTRAGE IN HEPARINIZED PLASMA

After storage at +4 °C, an overall decrease in enzymatic activity was observed in both matrices, more pronounced in serum than in heparinized plasma.

Conclusion

Ultimately, collection tube selection alongside storage conditions significantly impacts LDH measurement, emphasizing the necessity of strict, standardized pre-analytical practices to limit variability and to guarantee accurately results.